Saatwik Sairaam Vasamsetti

Data Analyst · Data Engineer

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PROFESSIONAL SUMMARY

Data-focused engineer who builds tested systems and checks every number back to its source: walk-forward ML back-testing in Python and scikit-learn, and a 60-table PostgreSQL + PostGIS backend verified by 1,603 assertions. Full Stack Web Developer Intern at Qaulium AI; co-founder of PathAhead. Seeking Data Analyst and Data Engineering internships.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, Kotlin, HTML, CSS
Data & Analytics: pandas, NumPy, scikit-learn, EDA, walk-forward back-testing, conformal prediction, Matplotlib, Power BI, Tableau, Excel
Databases & Data Eng.: PostgreSQL 16 + PostGIS, MySQL, Drizzle, schema & index design, migrations, concurrency control, SEC EDGAR ingestion
APIs, ML & Tools: FastAPI, REST, YOLO11, ONNX, AWS, Docker, Git, GitHub Actions CI, Vitest, Playwright

EXPERIENCE

Full Stack Web Developer Intern — Qaulium AI Private Limited *Jul 2026 – Present*

- Built the privacy engine of the Qualium Quantum Browser in TypeScript — ad and tracker blocking, cookie and storage isolation, fingerprint, WebRTC and DNS protection — as pure decision functions with no I/O, so every rule is testable without a browser.
- Covered it with 360 Vitest unit and integration tests across 22 files and 24 Playwright end-to-end tests, gated on typecheck and lint.

Co-Founder — PathAhead, career-guidance platform *Jun 2026 – Present*

- Built the live platform: 12 career roadmaps, a Career Builder and an AI counsellor grounded in PathAhead's own content, on two Netlify functions.

PROJECTS

Trade Research — Verifiable Research & Execution Platform *Aug – Sep 2026*

Stack: *Python, scikit-learn, pandas, FastAPI, PostgreSQL, React, MetaTrader 5, PyInstaller*

- Built a leakage-free walk-forward back-test of six scikit-learn models against naive and drift baselines, with conformal intervals, Monte Carlo paths and a cost-aware trading simulation.
- Made the numbers checkable: indicators computed in Python, financials from SEC EDGAR XBRL, and a verifier that traces every figure in a generated note back to the data — then reported the honest negative result.

RoadAssist Bharat — Offline-First Emergency Mobility Platform *Aug – Oct 2026*

Stack: *TypeScript, Fastify, PostgreSQL 16 + PostGIS, Drizzle, zod, Docker, GitHub Actions | github.com/saatwik-1157/roadassist-bharat*

- Built solo as a SWE4004 course project: an 86-route Fastify/TypeScript API over a 60-table PostgreSQL 16 + PostGIS schema with 151 indexes and 13 migrations.
- Verified by 1,603 assertions across six suites with zero failures, including 112 concurrency tests and 144 refused security attacks.
- Made races impossible in the database rather than by timing: row locks with expiry re-checked under the lock, a partial unique index guaranteeing "paid once", and a hash-chained, immutable audit log.

MachineSense AI — Explainable Industrial Intelligence *Smart India Hackathon 2026*

Stack: *Python, FastAPI, JavaScript, Canvas | github.com/saatwik-1157/machinesense-ai*

- Built a FastAPI predictive-maintenance prototype on simulated IoT sensor data: MAD z-score anomaly detection, 0-100 health scores explained per sensor, and remaining-useful-life forecasting.

CERTIFICATIONS

- **Google Analytics Certification** — Google, to May 2027
- **Data Fundamentals** — IBM SkillsBuild
- **Introduction to AWS** — AWS Training & Certification
- **Cloud Practitioner (CLF-C02) Domain 1 Review** — AWS
- **Data Engineer – Associate (DEA-C01) Exam Prep** — AWS
- **Exploring SAP Analytics Cloud** — SAP Learning
- **Becoming an SAP Data Architect** — SAP Learning
- **Architecting Agentic AI Business Solutions** — Microsoft Learn
- **Data Analytics Job Simulations** — Quantium and Deloitte, via Forage
- **GenAI-Powered Data Analytics Simulation** — Tata, via Forage

RECOGNITION & ACTIVITIES

- **Campus Ambassador** — SmartED Innovations, 2026
- **Smart India Hackathon 2026** — MachineSense AI
- **HackDevengers 1.0** — participant
- **SentinelAI** — Samsung Solve for Tomorrow & Ideas for India 2026
- **IIT Hyderabad Student Body Workshop** — Career & AI, Jun 2026

EDUCATION

VIT-AP University, Amaravati *2024 – 2029*
Integrated M.Tech — Computer Science & Engineering (in collaboration with Virtusa) *CGPA: 7.55 / 10*

SR Junior College *2022 – 2024*
Intermediate (Class XII, MPC) *94.8%*

DAV Public School
Secondary School (Class X) *81.17%*